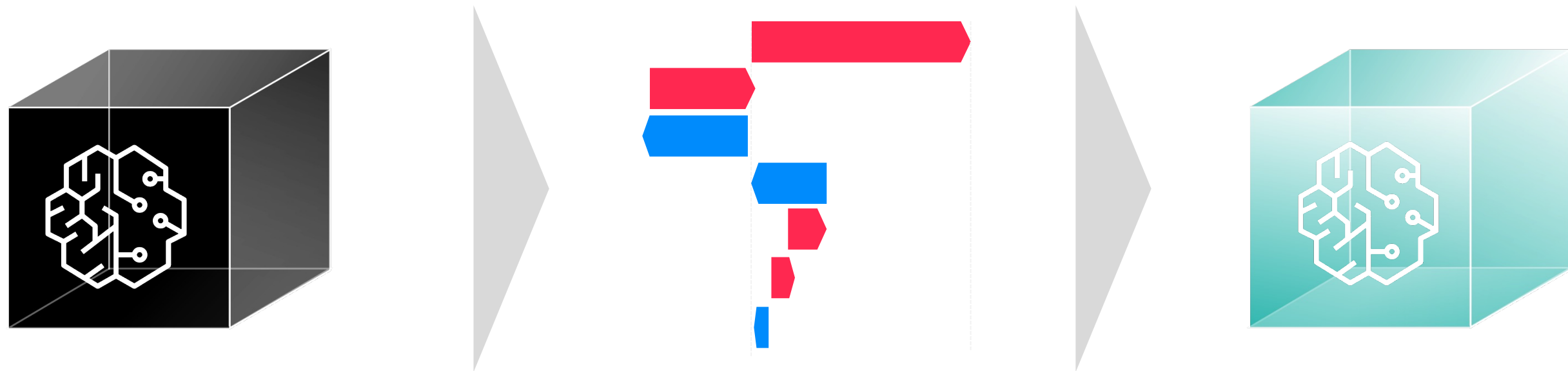
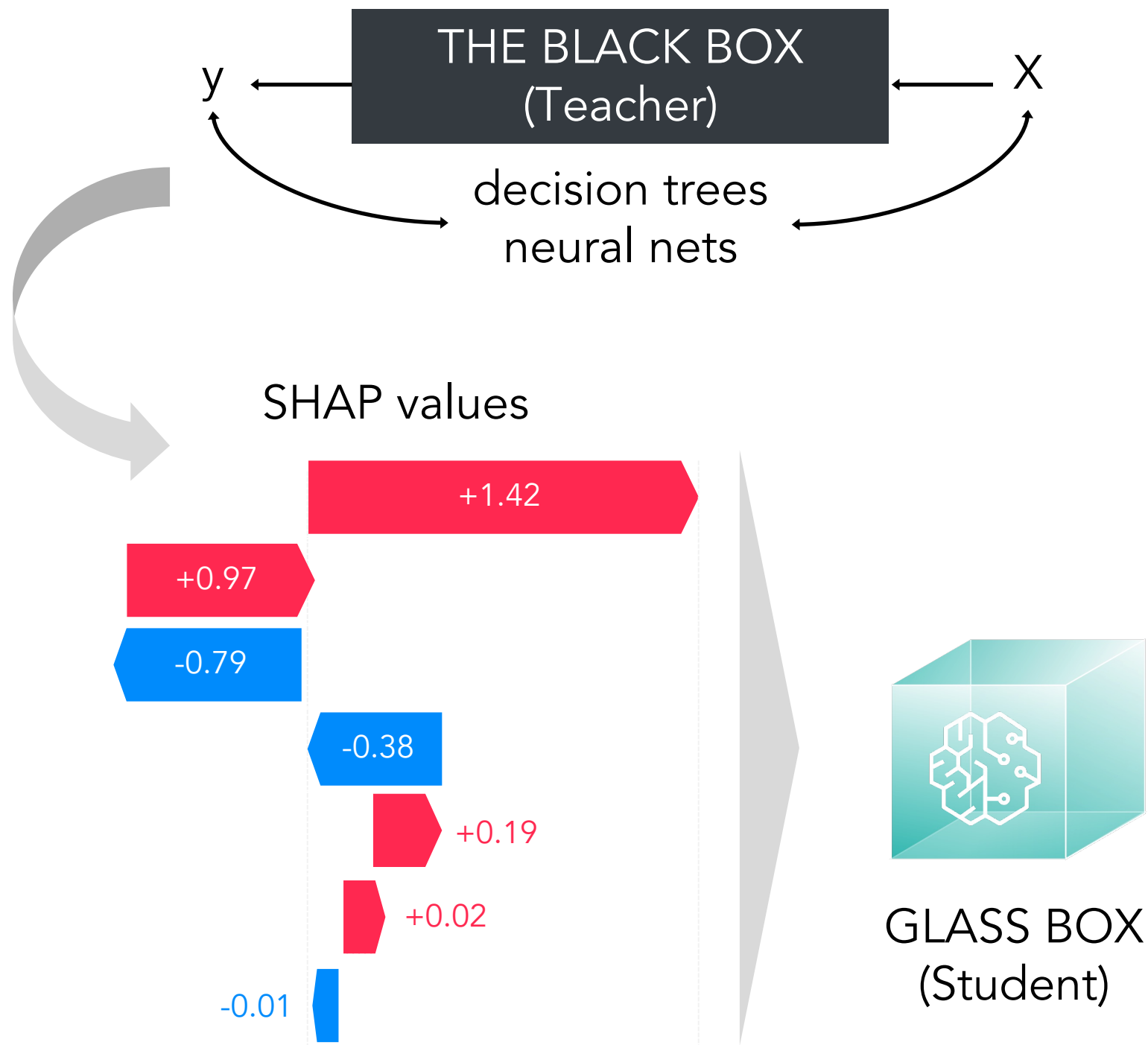


Model Distillation with SHAP Values



Model Distillation with SHAP Values

From Black Box to Glass Box



SHAP (SHapley Additive exPlanations) values allow us to decompose predictions of white-box and black-box machine learning (ML) models into additive scores that explain the outcome.

For more complex ML models, SHAP values leverage interactions between features, allowing us to represent complex relationships in a **human-understandable form**.

Knowing the SHAP values of model predictions allows us to perform distillation: building a simpler model (the student) that **approximates the teacher's predictive behavior**.

This deck explores two approaches¹:

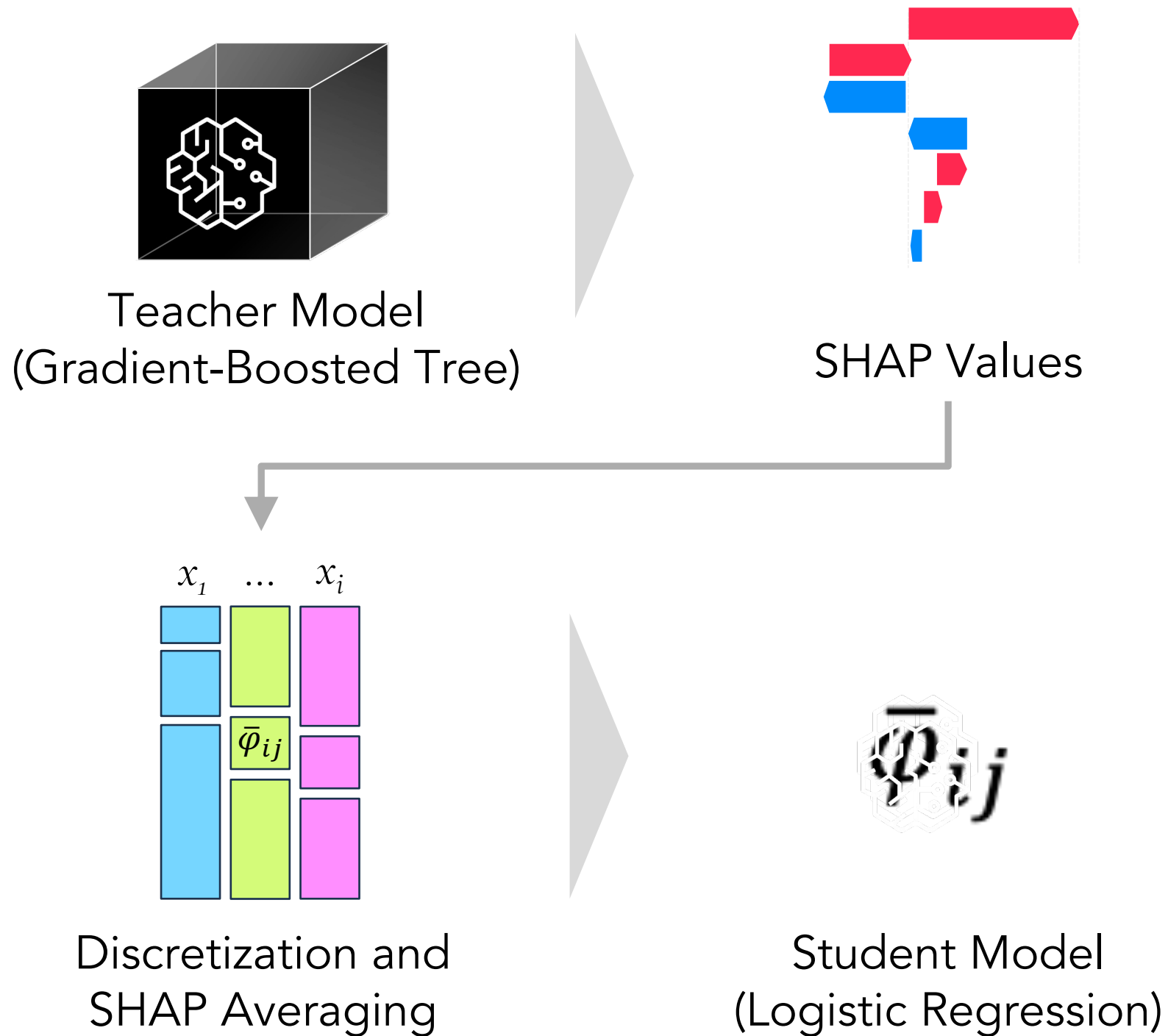
- A Linear model with binning;
- A Generalized Additive Model (GAM).

The new **glass box model** can be used for inference without requiring the original **black box model**.

1. These materials will become part of a chapter dedicated to SHAP values in Credit Risk, along with SHAP scorecards and other practical applications.

Model Distillation with SHAP Values

Approach 1: Binned SHAP Values with Logistic Regression



With this approach, we bin raw features and replace each bin with its average SHAP contribution.

This is conceptually similar to Weight of Evidence (WOE) used in scorecards. Because SHAP values are often expressed in log-odds units, this naturally produces a scorecard-like model structure.

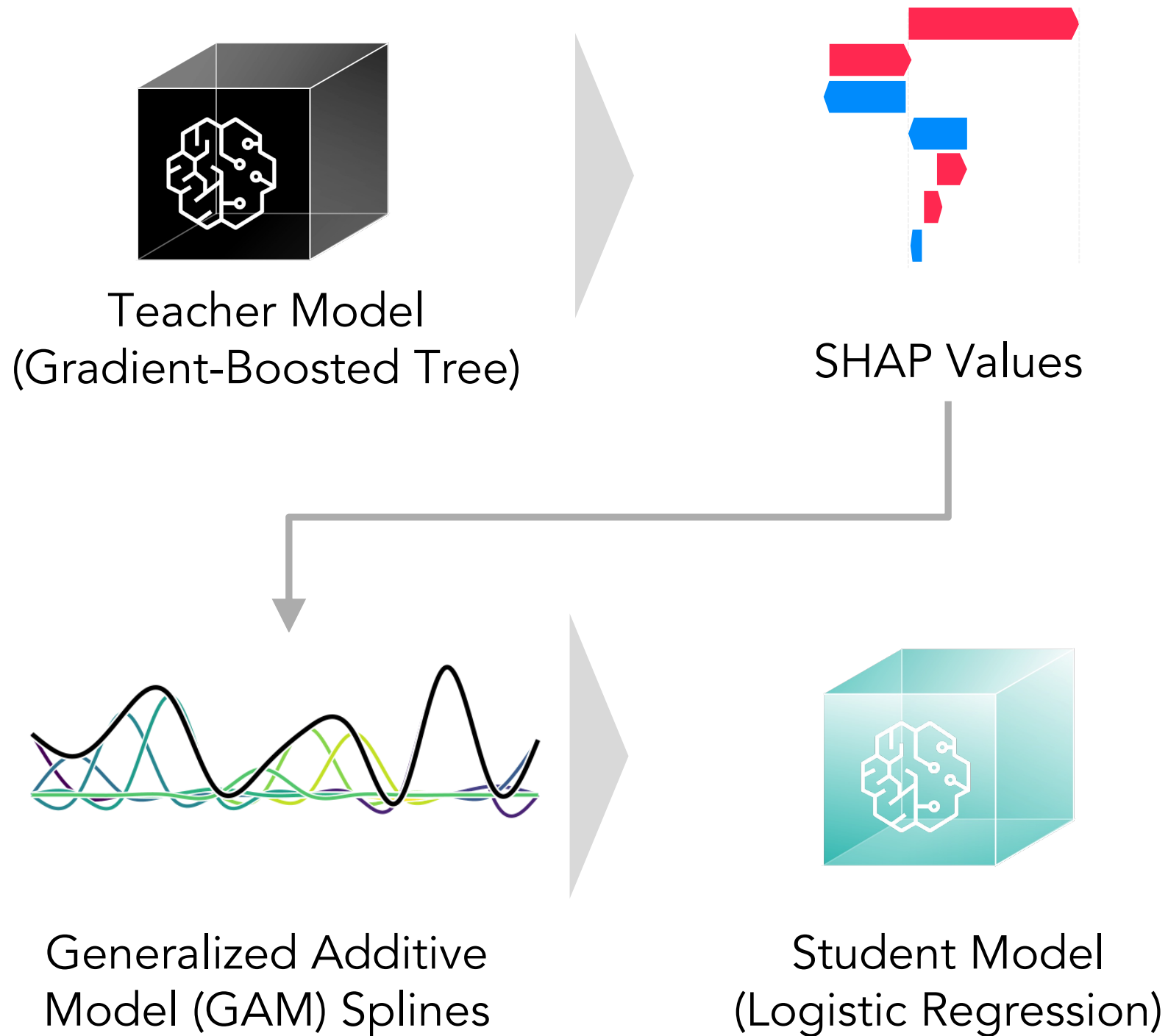
We can fit this pipeline as follows:

1. Train the **Teacher model** (e.g., Gradient-Boosted Decision Tree algorithm).
2. Compute **SHAP values** on the training data.
3. Discretize features into bins and compute average SHAP value for each bin
4. Replace each feature with mapped average SHAP value and **optionally create interactions**.
5. Train the **Student model** (Logistic Regression) on the transformed features.

This pipeline distills a complex model into an **interpretable logistic regression scorecard** using SHAP values.

Model Distillation with SHAP Values

Approach 2: GAM Splines with Logistic Regression



With this approach, we fit a Generalized Additive Model (GAM) for each feature so that each smooth function $g_j(x_j)$ approximates its SHAP contribution ϕ_{ij} . We then use these transformed features, optionally expanded with pairwise interactions, as inputs to a logistic regression model.

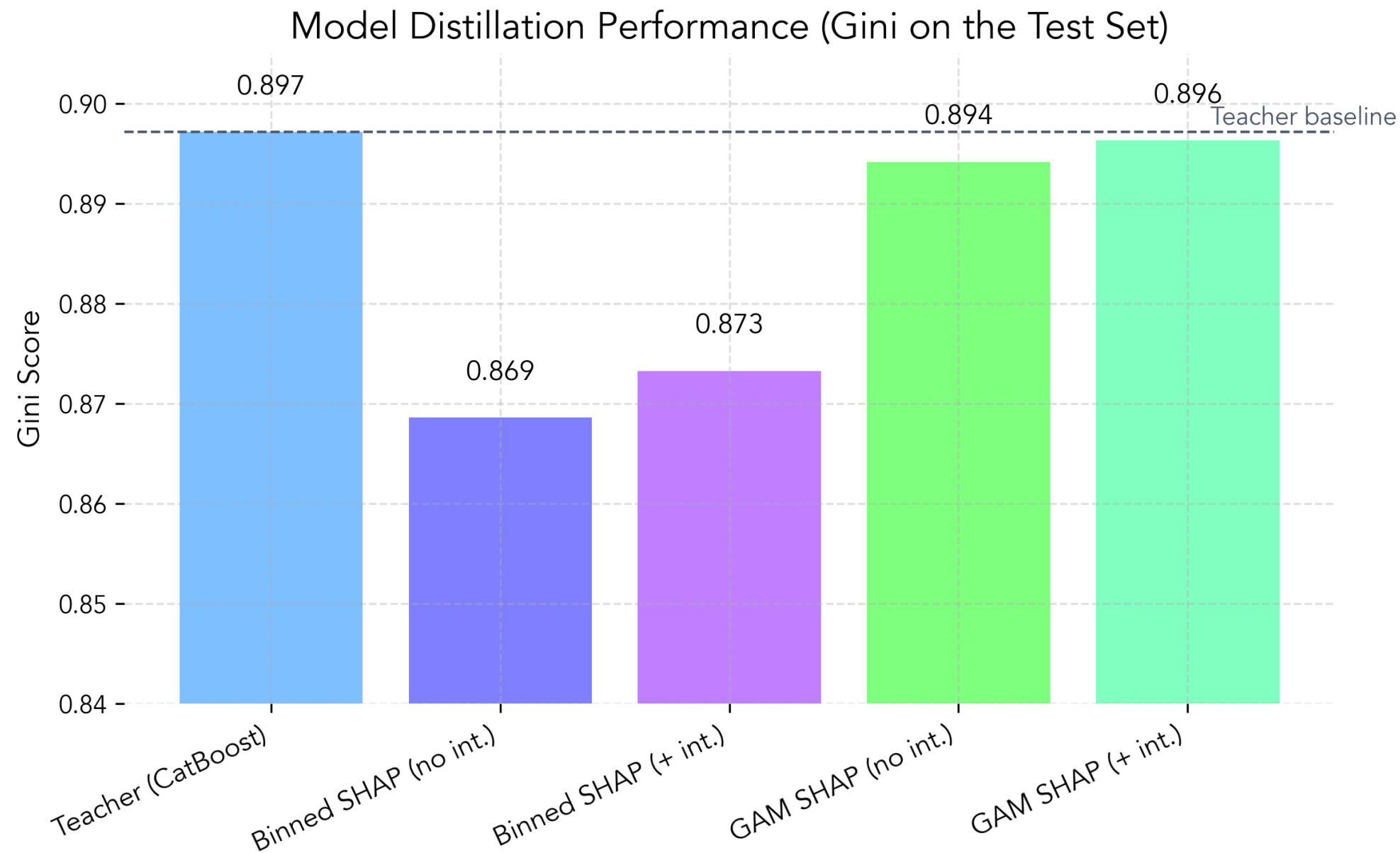
We can fit this pipeline as follows:

1. Train the **Teacher model** (e.g., Gradient-Boosted Decision Tree algorithm).
2. Compute **SHAP values** on the training data.
3. Fit a **GAM** $g_j(x_j) \approx \phi_{ij}$ for each feature.
4. Transform data using these GAM predictions and create optional **interaction terms**.
5. Train the **Student model** (Logistic Regression) on the transformed features.

This pipeline distills the nonlinear behavior of the teacher model into a GAM-enhanced logistic regression, preserving the teacher's predictive power while remaining interpretable.

Model Distillation with SHAP Values

Empirical Comparison of the Two Approaches



Our baseline is a CatBoost teacher trained on a credit risk dataset achieving a Gini score of around 0.896 on the test set.

When we distill it into a logistic regression using binned SHAP values, we retain around 96%–97% of the teacher’s discriminatory power.

SHAP-GAM distillation preserves almost all of the teacher’s performance, achieving up to 0.896 Gini, while keeping the model fully transparent and deployable as a scorecard.

Both SHAP-binned and SHAP-GAM students approximate the teacher’s performance quite well, with GAM-based distillation achieving a 99.9% retention rate of the teacher’s performance while remaining interpretable.